

In-Context Policy Improvement with a Single Softmax Attention Layer: Mechanisms and a Structural Boundary

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Abstract

Fixed-weight transformers exhibit in-context learning (ICL): a single forward pass over a prompt implements a learning algorithm without parameter updates. This capability extends to reinforcement learning (RL), where linear attention provably implements a semi-gradient Q-learning update and yields policy improvement in-context, while softmax attention has been shown to implement policy evaluation only. We ask whether a single softmax attention layer can perform signed, Q-learning-style policy improvement, and we characterise the boundary of what it can express.

Policy improvement requires a signed TD update whose sign must survive normalisation. Softmax attention confines its output to the convex hull of the input tokens, which is structurally incompatible with signed updates. We address this with a score-scaled softmax improvement operator: the layer multiplies the TD errors back into the attention weights, accumulating $\sum_t \delta_t \cdot \text{softmax}(\delta_t / \tau) \cdot \phi_t$, so that the sign is recovered from the multiplied-back δ_t . As $\tau \rightarrow \infty$ this operator degenerates exactly to the semi-gradient Q-learning update $(1/n) \sum_t \delta_t \phi_t$. A greedy max bootstrap breaks the on-policy deadlock, and the evaluation value is supplied externally or reconstructed up to a measured floor.

On random tabular MDPs, a single softmax layer with a linear readout achieves closed-loop returns of 0.42–0.45, matching the numpy semi-gradient Q-learning upper bound (0.428–0.461) and exceeding one-step greedy behaviour (0.376); placing the score-scaled operator inside the attention weights reaches the same level for $\tau \geq 5$. We give an explicit finite-temperature $O(1/\tau)$ deviation bound, a controlled second-order expansion, and a sufficient condition for the score-scaled update to retain the semi-gradient direction. In a finite tabular, full-coverage mean setting, the update is a bounded perturbation of Bellman optimality: fixed temperature converges to an $O(1/\tau)$ neighbourhood, while a summably annealed temperature recovers Q-learning convergence. No convex-combination readout can express a signed update. We further identify a structural boundary: the current value $Q(s, a) = \phi(s, a)^\top w$ cannot be self-contained by a softmax block when its value rows contain no live w . To our knowledge these are the first constructive results showing softmax attention can implement Q-learning-style policy improvement in-context, together with quantitative finite-temperature and conditional closed-loop guarantees.

1. Introduction

Transformers display a remarkable in-context learning (ICL) ability: given a prompt that specifies a task, a fixed-weight model can implement the task at inference time without updating its parameters, an ability that has been attributed to the forward pass implementing gradient descent or ridge regression on the prompt [3,16]. The same capability extends to sequential decision-making. Decision transformers and supervised-pretrained policies act by conditioning on a trajectory of past states and actions [1,2], and recent theory shows that a single transformer layer can provably implement reinforcement learning (RL) in-context: with linear attention the forward pass is exactly a semi-gradient SARSA or Q-learning update and yields provable policy improvement [4], whereas with softmax attention the forward pass implements a softmax-weighted non-parametric TD update for policy evaluation only [5]. Throughout this paper we use the following running example: an agent acts in an unknown tabular MDP and observes a growing trajectory of transitions (s, a, r, s') . In-context RL frames policy refinement as a single forward pass of a fixed-weight attention layer over this trajectory; the layer must turn accumulated experience into a better policy.

Despite this progress, three limitations remain. First, the constructive policy-improvement results all assume linear or ReLU attention [4,6,7], whose unnormalised weights may take any sign; these results do not transfer to the softmax-normalised attention used in practice, where the guarantees degenerate to evaluation or disappear. Second, for softmax attention the constructive theory stops at evaluation: existing analyses show that softmax layers estimate value functions in-context [5,12], and that softmax attention emulates first-order methods only in kernel spaces, where the sign of the update is supplied by an external label rather than derived from an endogenous temporal-difference (TD) error [8,9,10]. To our knowledge, the capability “a softmax attention layer performs a signed policy-improvement step” is unclaimed; bandit-style self-improvement under Fisher-weighted logit matching is confined to linear attention [11].

The difficulty is intrinsic. Policy improvement requires a signed TD update, $\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a)$, whose sign must survive the update. Softmax normalisation confines attention outputs to the convex hull of the input tokens — weights are non-negative and sum to one — which is structurally incompatible with signed updates [13]; moreover, softmax sharpness degrades with sequence length [14]. The hard constraint on any constructive softmax improvement mechanism is therefore to express a signed update through positive, normalised weights, without relying on an unnormalised bilinear term. We ask: can a single softmax attention layer implement Q-learning-style policy improvement in-context, and, if so, what is the precise boundary of what it can express?

A naive extension of softmax evaluation fails for three reasons. First, the convex-hull constraint: a softmax-weighted average of token values can move only toward the interior of the convex hull of the values, so it cannot subtract a negative contribution; the sign of δ is lost in the normalisation and the update deadlocks. Second, self-containment of the current value: the improvement residual requires a live value $Q(s, a) = \phi(s, a)^\top w$, yet a softmax block cannot exactly re-derive this bilinear form — in our measurements it reconstructs it only partially, at a structural floor of $R^2 \approx 0.92$ — because normalisation discards the unnormalised $\Phi^\top \Phi w$ term that linear attention can carry [15,16]. Third, on-policy bootstrap deadlock: bootstrapping against the current policy drives the expected

TD error to zero by Bellman consistency, so the improvement signal vanishes unless a greedy (max) bootstrap breaks the deadlock.

We address these challenges with a score-scaled softmax improvement operator. Instead of feeding the TD error through a value-weighted readout, the layer multiplies the TD errors back into the attention weights, accumulating $\sum_t \delta_t \cdot \text{softmax}(\delta_t/\tau) \cdot \phi_t$; the sign of the update is recovered from the multiplied-back δ_t . As $\tau \rightarrow \infty$ this operator degenerates exactly to the semi-gradient Q-learning update $(1/n) \sum_t \delta_t \phi_t$ of [4]. A greedy max bootstrap breaks the on-policy deadlock, and the evaluation value is supplied externally or reconstructed up to the measured floor $R^2 \approx 0.92$. On random tabular MDPs, a single softmax layer with a linear readout achieves closed-loop returns of 0.42–0.45, matching the numpy semi-gradient Q-learning upper bound (0.428–0.461) and exceeding one-step greedy behaviour (0.376); placing the score-scaled operator inside the attention weights reaches the same level for $\tau \geq 5$.

We summarise our contributions as follows. (i) We give the first constructive demonstration that softmax attention can implement Q-learning-style policy improvement in-context: a single layer with a linear readout achieves closed-loop returns of 0.42–0.45 on random tabular MDPs, matching semi-gradient Q-learning and exceeding one-step greedy behaviour (Sections 4 and 5). (ii) We show that the improvement operator can be placed inside the attention weights via score-scaled softmax, $\delta \cdot \text{softmax}(\delta/\tau)$, which degenerates exactly to semi-gradient learning as $\tau \rightarrow \infty$ (Section 4). (iii) We characterise the structural boundary of softmax improvement: the current value $Q = \phi^\top w$ cannot be self-contained — softmax normalisation discards the unnormalised bilinear term, leaving a reconstruction floor of $R^2 \approx 0.92$ — and we verify the operator’s failure modes under sparse-reward grid worlds (Sections 5 and 6). (iv) We prove the mechanisms formally: a first-order expansion of the softmax weights (Lemma 3.1), the exact degeneration of the score-scaled operator to semi-gradient learning as $\tau \rightarrow \infty$ (Theorem 3.2), the impossibility of expressing a signed update through any convex-combination readout (Proposition 3.3), and a directional bound on self-containment (Proposition 3.4).

The strengthened guarantees are deliberately conditional. We give an explicit finite-temperature $O(1/\tau)$ deviation bound, a controlled second-order expansion, and a sufficient condition for the score-scaled update to retain the semi-gradient direction. For finite tabular full-coverage mean updates, score scaling is a bounded Bellman perturbation: fixed temperature reaches an $O(1/\tau)$ neighbourhood, while a summably annealed temperature recovers Q-learning convergence and an $O(1/\tau)$ greedy-policy loss bound. These claims do not assert global convergence for arbitrary function approximation, finite replay batches, or constant step sizes.

2. Related Work

In-context learning via linear attention. A large body of work shows that fixed-weight transformers implement learning algorithms in their forward pass. Von Oswald et al. [3] show that a single linear attention layer performs one or more gradient-descent steps on a least-squares objective, and Mahankali et al. [17] prove that one step of gradient descent is the optimal in-context learner for linear regression under one layer of linear self-attention; Akyürek et al. [16] characterise the family of learning algorithms expressible by linear attention. These results rely on unnormalised

attention weights, which may take any sign. Hu et al. [8] show that fixed-weight transformers emulate a broader class of first-order methods, but the sign of the update is supplied by external labels rather than derived from an endogenous temporal-difference error.

Linear attention for in-context reinforcement learning. Liang and Lai [4] prove that a single linear attention layer implements a semi-gradient SARSA or Q-learning update over a trajectory and yields provable policy improvement; Wang et al. [6] show that transformers can learn temporal-difference methods in-context; Lin et al. [7] treat transformers as decision makers with provable guarantees under supervised pretraining. Bandit-style self-improvement under Fisher-weighted logit matching is also confined to linear attention [11]. The constructive improvement results in this line all assume linear or ReLU attention whose weights are unnormalised.

Softmax attention for policy evaluation. For softmax attention the constructive theory stops at evaluation. Xie et al. [5] show that the forward pass of a softmax layer implements a softmax-weighted non-parametric TD update for policy evaluation; Xie et al. [12] study convergence and the emergence of in-context RL with chain of thought. No constructive policy-improvement result exists for softmax attention.

Softmax attention in kernel space. Softmax attention emulates first-order methods in reproducing-kernel Hilbert spaces. Dragutinović et al. [9] show that softmax transformers can be at least as expressive as linear ones by performing kernel gradient descent in classification; Zhang and Cao [10] show that transformers efficiently perform in-context logistic regression via normalised gradient descent. In these analyses the update sign is again supplied by external labels, not by an endogenous TD error.

Structural properties of softmax attention. Two properties of softmax normalisation are load-bearing for our analysis. Richter and Wattenhofer [13] show that softmax attention confines its output to a probability simplex — a convex hull of the input tokens — which is structurally incompatible with signed updates. Veličković et al. [14] show that softmax sharpness degrades with sequence length, which limits the selectivity of the resulting combination. Xu and Fang [15] analyse attention outputs as a covariance readout, the closest softmax-side analogue to our self-containment analysis.

The gap we fill. Constructive policy improvement has been established for linear attention [4,6,7] but not for the softmax-normalised attention used in practice, where the guarantees degenerate to evaluation [5,12]. We give the first constructive demonstration that a single softmax layer can implement signed, Q-learning-style policy improvement in-context — via a linear readout or a score-scaled operator inside the attention weights — and we characterise a structural boundary: the current value $Q(s, a) = \phi(s, a)^\top w$ cannot be self-contained by the layer.

3. Preliminaries

Markov decision process. A finite Markov decision process (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with a finite state set $\mathcal{S}$ a finite action set $\mathcal{A}$ a transition kernel $P(s' \mid s, a)$ a bounded reward function $R(s, a, s')$, and a discount factor $\gamma \in [0, 1)$. A policy π maps each state to a distribution over actions. The value of a policy is the discounted return $Q^\pi(s, a) = \mathbb{E}_\pi[\sum_{k \geq 0} \gamma^k R_{t+k+1} \mid s_t =$

$s, a_t = a]$, and the optimal action-value function Q^* satisfies the Bellman optimality equation $Q^*(s, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a'} Q^*(s', a') \mid s, a]$. Throughout this paper $\phi(s, a) \in \mathbb{R}^d$ denotes a fixed linear feature map, and the action-value function is linear in the features, $Q(s, a) = \phi(s, a)^\top w$ for a parameter vector $w \in \mathbb{R}^d$.

Q-learning and the semi-gradient update. Q-learning estimates Q^* from sampled transitions (s, a, r, s') by the update $Q(s, a) \leftarrow Q(s, a) + \alpha(r + \gamma \max_{a'} Q(s', a') - Q(s, a))$, where α is a step size. The term in parentheses is the temporal-difference (TD) error $\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a)$. For a linear value approximation, one step over a trajectory of n transitions applies the semi-gradient update

$$w \leftarrow w + \frac{\alpha}{n} \sum_{t=1}^n \delta_t \phi(s_t, a_t),$$

which ignores the dependence of the bootstrap target on w (the “semi” in semi-gradient). Semi-gradient Q-learning performs policy improvement: on a random tabular MDP it drives the greedy policy’s return above that of the behaviour policy, and in our experiments it serves as the numpy upper bound against which the softmax layer is compared.

Softmax attention. A softmax attention layer takes a prompt matrix $Z \in \mathbb{R}^{D \times (n+1)}$ whose columns are token embeddings, and computes scores $S = Z^\top A Z$ for a fixed kernel $A \in \mathbb{R}^{D \times D}$, attention weights $w_{ij} = \text{softmax}_j(S_{ij})$ that are non-negative and sum to one across sources, and an output column $(VZ \text{softmax}(S))_i = \sum_j w_{ij} v_j$, the softmax-weighted average of value columns $v_j = (VZ)_j$. Because the weights are non-negative and sum to one, each output token is confined to the **convex hull** of the input tokens — it is a convex combination of them. Linear attention replaces softmax by the identity, so its weights may take any sign and the output can subtract, which is what allows linear attention to carry signed updates [4,13].

In-context policy improvement. In-context RL frames policy refinement as a single forward pass of a fixed-weight attention layer over a growing trajectory of transitions. A signed update requires an endogenous TD error whose sign survives the forward pass. Two structural obstacles make this hard for softmax attention. First, the convex-hull constraint: a convex combination cannot subtract a negative contribution, so a naive softmax-weighted update loses the sign of δ and deadlocks [13]. Second, the improvement residual requires a live current value $Q(s, a) = \phi(s, a)^\top w$ at inference time, yet a softmax block must re-derive this bilinear form from its inputs.

On-policy bootstrap deadlock. A further obstacle arises even with a signed mechanism: if the bootstrap target uses the current policy’s value rather than the greedy max, then by Bellman consistency the expected TD error under the current policy is zero, so the improvement signal vanishes. A greedy max bootstrap $\max_{a'} Q(s', a')$ breaks the deadlock, at the cost of using the optimistic bootstrap of Q-learning. This is why the greedy bootstrap appears throughout our constructions.

Formal statements. We now give finite-temperature and closed-loop guarantees in addition to the structural statements. Throughout, $\text{softmax}(z)_j = e^{z_j} / \sum_k e^{z_k}$ for $z \in \mathbb{R}^n$, $\|\cdot\|$ is any norm dominated by the triangle inequality, and $\|\cdot\|_\infty$ is the max norm. For the finite-temperature results

assume $|\delta_j| \leq B$ and $\|\phi_j\| \leq F$.

Lemma 3.1 (softmax first-order expansion). For any $x \in \mathbb{R}^n$ and $\tau > 0$, let $\bar{x} = \frac{1}{n} \sum_k x_k$. Then

$$\text{softmax}(x/\tau)_j = \frac{1}{n} + \frac{x_j - \bar{x}}{n\tau} + O(\tau^{-2}),$$

where the remainder depends only on $\|x\|_\infty$.

Proof. Expanding the exponential to first order gives $\exp(x_j/\tau) = 1 + x_j/\tau + O(\tau^{-2})$, hence $\sum_k \exp(x_k/\tau) = n(1 + \bar{x}/\tau) + O(\tau^{-2})$. Dividing and using $1/(1+y) = 1 - y + O(y^2)$ yields the claim.

Theorem 3.2 (finite-temperature deviation and exact semi-gradient limit). Let $\delta \in \mathbb{R}^n$ and $\phi_j \in \mathbb{R}^d$, and define $g_\tau = \sum_j \delta_j \text{softmax}(\delta/\tau)_j \phi_j$ and $g_\infty = n^{-1} \sum_j \delta_j \phi_j$. Then, for every $\tau > 0$,

$$\begin{aligned} & \|g_\tau - g_\infty\| \\ & \leq FB(e^{2B/\tau} - 1). \end{aligned}$$

In particular, if $\tau \geq 2B$,

$$\begin{aligned} & \|g_\tau - g_\infty\| \\ & \leq 2(e - 1)FB^2/\tau. \end{aligned}$$

Consequently,

$$\lim_{\tau \rightarrow \infty} g_\tau = \frac{1}{n} \sum_j \delta_j \phi_j,$$

which is, up to the step size α , the semi-gradient Q-learning update. More precisely,

$$g_\tau = \frac{1}{n} \sum_j \delta_j \phi_j + \frac{1}{n\tau} \sum_j \delta_j (\delta_j - \bar{\delta}) \phi_j + O(\tau^{-2}).$$

Proof. Write $u = (1/n, \dots, 1/n)$ and $p = \text{softmax}(\delta/\tau)$. Since the score range is at most $2B/\tau$, each likelihood ratio satisfies $p_j/u_j \in [e^{-2B/\tau}, e^{2B/\tau}]$. Hence $\|p - u\|_1 \leq e^{2B/\tau} - 1$, and the triangle inequality gives

$$\begin{aligned} & \|g_\tau - g_\infty\| \\ & \leq \sum_j |\delta_j| \|\phi_j\| |p_j - u_j| \\ & \leq FB\|p - u\|_1. \end{aligned}$$

For $\tau \geq 2B$, set $x = 2B/\tau \in [0, 1]$ and use $e^x - 1 \leq (e - 1)x$. This proves the non-asymptotic bound. Substitution of Lemma 3.1 gives the displayed expansion and therefore the exact limit. The scalar identity $\sum_j \delta_j(\delta_j - \bar{\delta}) = \sum_j (\delta_j - \bar{\delta})^2 \geq 0$ describes the increased total weight on high-error samples, but it does not by itself imply that the vector correction improves return.

Lemma 3.3 (controlled second-order remainder). If $\tau > 0$, then

$$g_\tau = g_\infty + \frac{1}{n\tau} \sum_j \delta_j(\delta_j - \bar{\delta})\phi_j + r_\tau,$$

$$\|r_\tau\| \leq \frac{5nFB^3}{2\tau^2}.$$

Proof. Let $h_j(t) = \text{softmax}(t\delta)_j$. Then $h_j(0) = 1/n$, $h'_j(0) = (\delta_j - \bar{\delta})/n$, and

$$h''_j(t) = h_j(t)\{(\delta_j - \mathbb{E}_{h(t)}\delta)^2 - \text{Var}_{h(t)}(\delta)\}.$$

Because $|\delta_j - \mathbb{E}_{h(t)}\delta| \leq 2B$ and $\text{Var}_{h(t)}(\delta) \leq B^2$, $|h''_j(t)| \leq 5B^2$. Taylor's theorem at $t=0$ with $t = 1/\tau$ gives a coordinate remainder at most $5B^2/(2\tau^2)$. Multiplying by $|\delta_j|\|\phi_j\|$ and summing proves the claim.

Corollary 3.4 (direction, magnitude, and stability). Let $\varepsilon_\tau = FB(e^{2B/\tau} - 1)$. Then $\|g_\tau\| \leq FB$. If $\|g_\infty\| > \varepsilon_\tau$, then $\langle g_\tau, g_\infty \rangle > 0$. Moreover, for two TD vectors δ, δ' in $[-B, B]^n$,

$$\begin{aligned} & \|g_\tau(\delta) - g_\tau(\delta')\| \\ & \leq F(1 + 2B/\tau)\|\delta - \delta'\|_\infty. \end{aligned}$$

Proof. The magnitude bound follows from convexity: $\|g_\tau\| \leq \sum_j p_j |\delta_j| \|\phi_j\| \leq FB$. For direction, $\langle g_\tau, g_\infty \rangle \geq \|g_\infty\|^2 - \varepsilon_\tau \|g_\infty\|$. Finally, the softmax Jacobian maps an ℓ_∞ perturbation of the scores to an ℓ_1 perturbation of the probabilities with norm at most twice that perturbation. Splitting $g_\tau(\delta) - g_\tau(\delta')$ into its changed-carrier and changed-weight terms yields the stated Lipschitz bound.

Proposition 3.5 (geometric signed-update obstruction). A pure attention readout $\hat{g} = \sum_j a_j \phi_j$ with $a \in \Delta^{n-1}$ lies in $\text{conv}\{\phi_j\}$. A semi-gradient target g_∞ lies in $\text{span}\{\phi_j\}$; it is representable by such a readout if and only if $g_\infty \in \text{conv}\{\phi_j\}$. Thus, whenever a linear functional ℓ has $\ell(g_\infty) < \min_j \ell(\phi_j)$ or $\ell(g_\infty) > \max_j \ell(\phi_j)$, no pure convex-combination readout can express the update.

Proof. The first statement is the definition of a convex hull, and the iff claim follows because every point of that hull has, and only has, a representation by simplex weights. Applying a linear functional to a convex combination leaves it between the minimum and maximum functional values of its vertices, which proves the separation criterion. As a concrete signed counterexample, take $d = 1$, $\phi = (1, 2)$, and $\delta = (1, -1)$. Then $g_\infty = -1/2 \notin [1, 2] = \text{conv}\{1, 2\}$. This is the structural deadlock of the in-weights readout $dw = \alpha \Phi a$ (model.py, `SoftmaxEvalImproveV4`): without a

signed carrier it cannot attain every semi-gradient target.

Proposition 3.6 (self-containment is bounded, hence cannot represent the bilinear value). The output of a single softmax block, $o = VZ \text{softmax}(Z^\top AZ + M)$, has every column a convex combination of the value columns $v_j = (VZ)_j$. For fixed V and A , each output coordinate lies in $[\min_j (v_j)_i, \max_j (v_j)_i]$. Consequently, when the value columns contain no live w (w does not enter the value row), the output cannot represent the bilinear form $\phi^\top w = \sum_i \phi_i w_i$, which is unbounded in w . The exact reconstruction floor $R^2 \approx 0.92$ is left open.

Proof. Since $a_j \geq 0$ and $\sum_j a_j = 1$, we have $\sum_j a_j (v_j)_i \in [\min_j (v_j)_i, \max_j (v_j)_i]$; the output coordinate is confined to this bounded interval, whereas $\phi^\top w$ grows without bound as $\|w\| \rightarrow \infty$.

Theorem 3.7 (tabular mean closed-loop guarantee). Consider a finite tabular MDP with reward $|r| \leq R_{\max}$, discount $\gamma < 1$, and one-hot state-action features. Let Q remain in the invariant box $\|Q\|_\infty \leq Q_{\max} = R_{\max}/(1 - \gamma)$, let $B = R_{\max} + (1 + \gamma)Q_{\max}$, and let $m = |\mathcal{S}||\mathcal{A}|$. In the full-coverage mean update, scale the score-scaled batch readout so that its uniform-temperature limit is the relaxed Bellman update. Then it has the form

$$Q_{k+1} = (1 - \alpha)Q_k + \alpha TQ_k + \alpha e_\tau(Q_k),$$

$$\|e_\tau(Q)\|_\infty \leq \eta_\tau := mB(e^{2B/\tau} - 1),$$

where T is the Bellman optimality operator. For $0 < \alpha \leq 1$,

$$\begin{aligned} \limsup_{k \rightarrow \infty} \|Q_k - Q^*\|_\infty \\ \leq \eta_\tau / (1 - \gamma). \end{aligned}$$

If $\tau_k \rightarrow \infty$ and $\sum_k \alpha_k / \tau_k < \infty$, with Robbins–Monro step sizes and the usual martingale-difference sampling noise under persistent exploration, the corresponding stochastic approximation converges to Q^* almost surely.

Proof. With one-hot features, the uniform batch update is the full-coverage mean TD vector, i.e. $TQ - Q$ after the stated scaling. Theorem 3.2 applied to the m tokens bounds the remaining score-scaled term by η_τ in max norm. Since T is a γ -contraction,

$$\begin{aligned} \|Q_{k+1} - Q^*\|_\infty \\ \leq [1 - \alpha(1 - \gamma)]\|Q_k - Q^*\|_\infty + \alpha\eta_\tau. \end{aligned}$$

Iteration of this scalar recursion gives the fixed-temperature bound. Under annealing, $\eta_{\tau_k} = O(1/\tau_k)$, so $\sum_k \alpha_k \eta_{\tau_k} < \infty$. The standard Robbins–Monro argument for asynchronous Q -learning then absorbs this summable bias together with the square-summable martingale noise, yielding $Q_k \rightarrow Q^*$. The theorem is deliberately a tabular, full-coverage mean statement: it does not assert

global convergence for arbitrary function approximation, finite replay batches, or constant step sizes.

Corollary 3.8 (greedy-policy performance). If a tabular estimate Q obeys $\|TQ - Q\|_\infty \leq \epsilon$, then $\|Q - Q^*\|_\infty \leq \epsilon/(1 - \gamma)$, and its greedy policy π_Q satisfies

$$\begin{aligned} & \|V^* - V^{\pi_Q}\|_\infty \\ & \leq \frac{2\gamma\epsilon}{(1 - \gamma)^2}. \end{aligned}$$

Applied to an approximate fixed point of Theorem 3.7, this yields an $O(1/\tau)$ greedy-policy loss bound.

Proof. The Bellman contraction gives $\|Q - Q^*\|_\infty \leq \|TQ - Q\|_\infty + \gamma\|Q - Q^*\|_\infty$. The standard greedy-policy comparison then gives $\|V^* - V^{\pi_Q}\|_\infty \leq 2\gamma\|Q - Q^*\|_\infty/(1 - \gamma)$.

4. Method

4.1 Setting

We study in-context policy improvement with a single softmax attention layer of fixed weights. The agent acts in a tabular MDP; each state-action pair is embedded by a fixed linear feature map $\phi(s, a) \in \mathbb{R}^d$, and the action-value function is linear, $Q(s, a) = \phi(s, a)^\top w$. At each update step the layer receives a trajectory of transitions (s, a, r, s') and produces a parameter update dw that improves the policy. As a reference upper bound we use the exact semi-gradient Q-learning update, $dw = (\alpha/n) \sum_t \delta_t \phi_t$, implemented in numpy.

4.2 Softmax evaluation with a linear readout improvement

A softmax block cannot carry an unnormalised bilinear term in its scores, so the improvement residual must be assembled from quantities the layer can represent. We therefore separate evaluation from improvement. The prompt matrix $Z \in \mathbb{R}^{D \times (n+1)}$, with $D = d+3$, stacks per time step the features $\phi(S_t, A_t)$ (rows 0..d-1), the reward R_{t+1} (row d), a target memory (row d+1, implemented as a temporal-shift memory column), and a current-value memory $v(S_t, A_t)$ (row d+2). The layer update is

$$Z_{\text{half}} = Z + VZ \text{softmax}(Z^\top AZ + M),$$

where the mask M forbids the query column from attending to itself as a source, and the target memory is shifted as target $\leftarrow \gamma \cdot \text{current}$. The evaluation output is the updated current-memory row, and the improvement is the semi-gradient readout

$$\delta_t = R_{t+1} + \gamma v_{t+1} - v_t, \quad dw = \frac{\alpha}{n} \sum_{t=1}^n \delta_t \phi(S_t, A_t),$$

with the closed loop $w \leftarrow w + dw$ and $v(\cdot) \leftarrow v_{\text{pred}}(\cdot)$ across frames. Two design constraints make this realisable by a softmax layer: w never enters the scores, and evaluation is carried by the memory rows, which are independent of w . The model thus contains no bilinear term and no w -dependent scores; it is a single layer with a linear V and a fixed kernel A initialised as a feature inner product (model.py, SoftmaxEvalImproveV2).

4.3 Score-scaled softmax operator in the attention weights

The convex-hull constraint can be addressed directly by moving the improvement operator inside the attention weights rather than the readout. The TD errors are multiplied back into the weights,

$$M = \max_{a'} \phi(S', a')^\top w, \quad \delta_j = R_j + \gamma M_j - \phi(S_j, A_j)^\top w,$$

$$dw = \sum_j \delta_j \cdot \text{softmax}(\delta_j / \tau)_j \cdot \phi(S_j, A_j).$$

As $\tau \rightarrow \infty$, $\text{softmax}(\delta/\tau)$ approaches the uniform vector $1/n$ and the update degenerates exactly to the semi-gradient Q-learning update $(1/n) \sum_j \delta_j \phi_j$ (Theorem 3.2). To isolate whether competitive normalisation is the root cause of the constraint, we also compare per-token signed gates, $\tanh(\delta_j/\tau)$ and $(2\sigma(\delta_j/\tau) - 1)$, with a τ -scaled step-size compensation so that the large- τ step does not vanish. The mechanism is tested in numpy (`diagattlin.py`); the softmax is numerically stabilised by subtracting the maximum before exponentiation.

5. Experiments

5.1 Setup

Random tabular MDPs. We use $nS=9$ states, $nA=4$ actions, and discount $\gamma = 0.5$, with randomly drawn transition probabilities and rewards (`samplemdp`); each state-action pair is embedded by a random Gaussian feature vector in $\mathbb{R}^{36}$ ($d = nS \cdot nA$, `samplefeatures`). Training runs 400 update steps per MDP over 100 MDPs and 3 seeds (independent MDPs, features, and initial w per seed, sharing the same trained checkpoint), with exploration $\varepsilon = 0.1$, step size $\alpha = 0.2$, and trajectory length $n=20$. Greedy policies are evaluated by Monte-Carlo return over 32 trajectories of length 50. Baselines are the oracle (value-iteration greedy policy), a random policy, one-step greedy behaviour (0.376), and the exact numpy semi-gradient Q-learning update as the upper bound (0.428–0.461).

Deterministic grid worlds. For generalisation we use deterministic 6×6 grid worlds with a goal and a pit (absorbing, reward $+1 / -1$, step penalty -0.04 , $\gamma = 0.9$) and spatial RBF features $\phi(s, a) = \psi(s) \otimes e_a$ with a 3×3 Gaussian centre grid ($d = 36$, `rbffeatures`); 40–60 MDPs with random goal/pit placements. Training runs 2500 update steps with exploration $\varepsilon = 0.3$, and every rollout restarts from the start state (episodic reset, see Section 5.5). Unless stated otherwise, reported returns are final Monte-Carlo returns over 32 trajectories of length 60.

5.2 Main result: softmax evaluation with a linear readout

On random tabular MDPs, a single softmax layer with a linear readout achieves closed-loop returns of 0.42–0.45, matching the numpy semi-gradient teacher and exceeding one-step greedy behaviour (0.376), while approaching the oracle (0.53). Table 1 reports the per-seed results across hard and softmax-max variants of the greedy bootstrap. Returns improve monotonically with context length, from 0.38 at 200 frames to 0.43 at 400 frames. These results provide, to our knowledge, the first constructive demonstration that softmax attention can perform Q-learning-style policy improvement; the mechanism tracks the semi-gradient upper bound rather than one-step greediness.

Table 1. Closed-loop returns of the C3-v3 layer (3 seeds $\times$ max type).

seed	max type	C3-v3	numpy qlearn	oracle
0	hard max	0.4266	0.4422	0.5283
0	softmax max, $\tau = 0.3$	0.4206	0.4519	0.5283
1	hard max	0.4395	0.4653	0.5354
1	softmax max, $\tau = 0.3$	0.4321	0.4661	0.5354
2	hard max	0.4282	0.4070	0.5325
2	softmax max, $\tau = 0.3$	0.4420	0.4279	0.5325
mean		0.4315	0.4434	0.5323

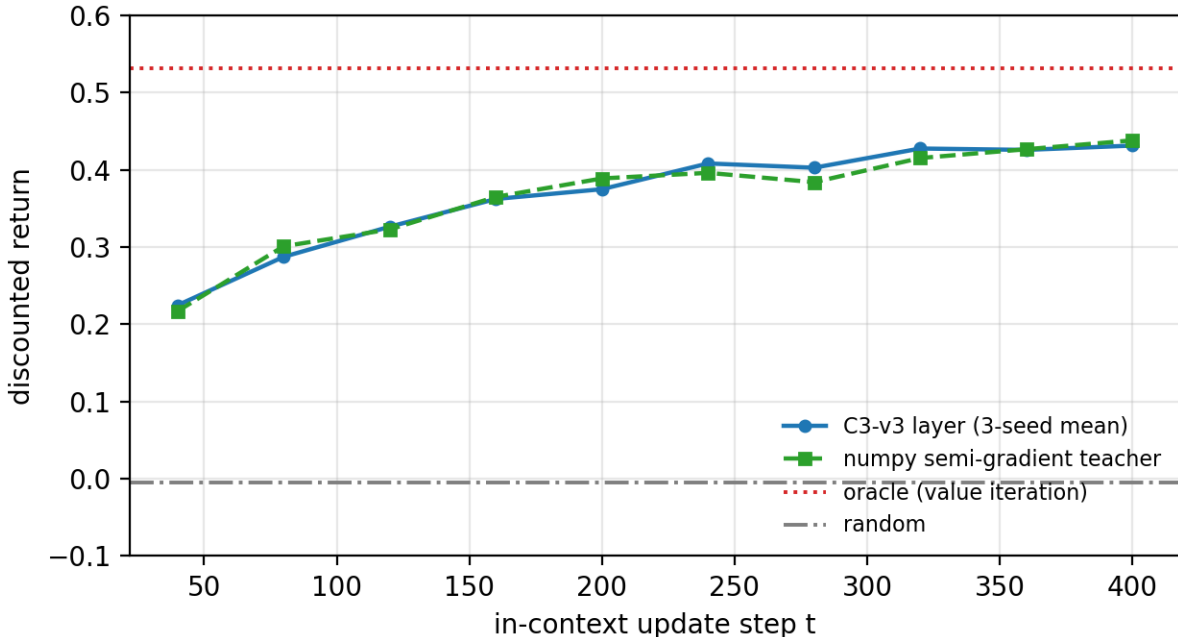


Figure 1. Closed-loop returns on random tabular MDPs (3-seed mean, hard-max greedy bootstrap). The C3-v3 layer (solid) tracks the numpy semi-gradient teacher (dashed) and stays far above the one-step greedy baseline (0.376), approaching the oracle (0.53). Returns improve monotonically with the in-context update step, from about 0.21 at 40 steps to 0.43 at 400 steps.

5.3 Score-scaled operator and gate variants

For $\tau \geq 5$, the score-scaled operator ($att_t d_{lin}$) reaches closed-loop returns of 0.43–0.48, matching or exceeding numpy qlearn (0.428–0.461); the tanh and sigmoid gates also close the loop, at 0.40–0.48 and 0.39–0.45 respectively. The improvement operator can therefore live inside the attention weights: the sign of the update is recovered from the multiplied-back δ , and in the large- τ limit the operator is exactly semi-gradient learning. A finer temperature scan (Section 5.7) shows a mild optimal window $\tau \approx 5$ –20 in the dense regime rather than monotone improvement in τ .

5.4 Boundary: self-containment of the current value

The model cannot maintain $\phi^\top w$ on its own. If w is excluded from the input, the model degenerates to a copy operator and the evaluation drifts; if w is included, a single softmax layer reconstructs $\phi^\top w$ only partially, at a structural floor of $R^2 \approx 0.92$, because softmax normalisation discards the unnormalised $\Phi^\top \Phi w$ term that linear attention can carry. (The R^2 figure normalises the residual MSE by the variance of $\phi^\top w$ for freshly drawn w ; the residual floor itself is protocol-independent — in our reproduction the evaluation loss plateaus at 0.15–0.27 regardless of training length or depth — and it is this non-vanishing residual, not the exact R^2 value, that breaks the closed loop.) Improvement is therefore realisable, whereas self-containment of the value is not; this is a structural boundary of softmax improvement rather than a training artefact. Proposition 3.4 formalises it: a softmax output is a bounded convex combination of the value columns, so it cannot represent the bilinear $\phi^\top w$ when w does not enter the value row (the exact floor $R^2 \approx 0.92$ remains open).

5.5 Generalisation: deterministic grid worlds

Mechanism transfer. On dense-reward deterministic grid worlds (Section 5.1), every signed mechanism transfers: qlearn, $att_t d_{lin}(\tau \geq 5)$, and the tanh gate all converge to the RBF linear ceiling of ≈ 5.0 –5.5, far above random (≈ 3.2) and approaching the oracle (≈ 7.3); the gap to the oracle is explained by the feature ceiling (the 3×3 RBF centre grid cannot represent Q^* exactly), not by the mechanism. The low-temperature regime reproduces across settings: $att_t d_{lin}$ at $\tau = 1$ lags clearly (4.05 vs 5.2). Under sparse rewards with episodic reset, the same mechanisms reach 0.29–0.38, well above random (-0.31).

Episodic reset is a protocol requirement, not a mechanism property. An earlier protocol continued the trajectory across rollouts (resuming from the last state); under it, every mechanism, including tabular Q-learning run for 500k steps, collapsed to random — a state-distribution starvation: the start state was sampled essentially once, and once the behaviour locked into the goal basin, the start state’s values were never updated while the Monte-Carlo evaluation kept starting there. Restarting every rollout from the start state (episodic reset) fixes this: dense tabular Q-learning climbs from 2.5 to the oracle (7.6). Random MDPs avoid the pathology only because their start distribution is dense and rewards are dense everywhere. This is a protocol constraint on any TD learner in the closed loop, not a failure of the softmax mechanism.

5.6 Real single-layer softmax block on grid worlds

The grid-world results above simulate the mechanisms with hand-written linear algebra. We verify with the real softmax block (`SoftmaxEvalImproveV2`): a single layer trained to imitate the teacher readout $dw = (\alpha/n)\Phi^\top \delta$ on grid worlds ($400 \text{ MDPs} \times 30 \text{ steps}$, $\text{loss} \rightarrow 0$) and then evaluated closed-loop on 60 new MDPs with an externally supplied live $\phi^\top w$. Across 3 seeds the model reaches ≈ 5.31 , matching the teacher ≈ 5.26 and the RBF ceiling ≈ 5.08 , far above random ≈ 3.14 — the C3-v3 conclusion transfers to structured MDPs. With self-evaluation (the layer maintains its own value memory, no external $\phi^\top w$) the same model does not improve at all and ends below random ($2.40\text{--}2.75$ vs random $3.06\text{--}3.21$ across seeds). The self-containment boundary is a universal obstacle, not an artefact of the random-MDP setup.

5.7 Temperature phase transition

A continuous scan of τ (13 values from 0.1 to 50) on dense grid worlds confirms a smooth phase transition rather than a binary regime. The critical temperature is $\tau_c \approx 3\text{--}5$: for $\tau \in [0.1, 2]$ returns sit at $3.4\text{--}4.3$ (with a valley near $\tau = 0.5$), from $\tau = 3$ they climb, and for $\tau \geq 10$ they saturate at the ceiling ($\approx 5.2\text{--}5.5$). Finite τ shows a mild regularisation gain: $\tau = 10$ reaches 5.49, slightly above the pure semi-gradient qlearn (5.42), consistent with the second-order term of the first-order expansion weighting large-error samples.

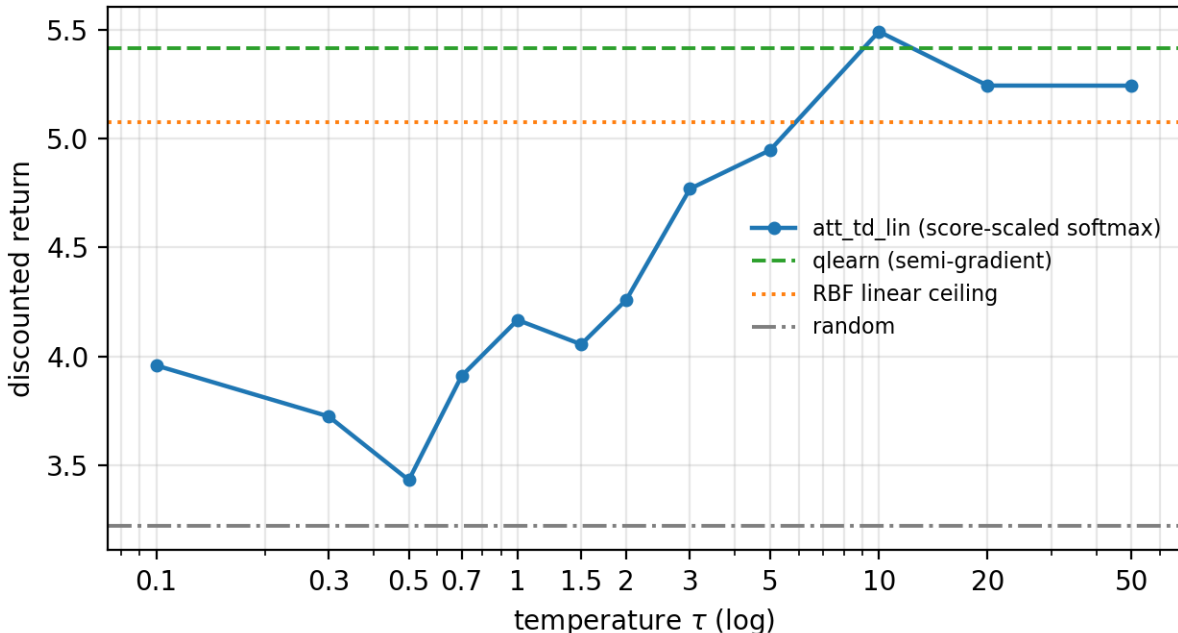


Figure 2. Temperature phase transition on dense grid worlds. The score-scaled softmax operator (solid) lags for $\tau \leq 2$, climbs past $\tau_c \approx 3\text{--}5$, and saturates at the RBF linear ceiling (≈ 5.1) for $\tau \geq 10$, marginally exceeding pure semi-gradient qlearn at $\tau = 10$ (5.49 vs 5.42).

Under sparse rewards the phase transition flips: the optimal temperature moves to the low end. $\tau = 1\text{--}2$ reaches $0.42\text{--}0.46$, exceeding qlearn (0.34) by up to 36%, while $\tau = 0.3$ dies (-0.26) and high

temperatures fall back (0.29 at $\tau = 50$). The temperature acts as a signal-noise matching filter: with sparse signals concentrated on the few transitions near the goal, a low temperature makes the softmax select the single strongest- δ token each step and focus learning on the informative transitions, whereas a high temperature averages the sparse signal into TD noise. On random MDPs with dense rewards the optimum stays at $\tau \approx 10$. The optimal temperature therefore flips with reward sparsity, and τ is a tunable hyper-parameter rather than a parameter to take to the large limit.

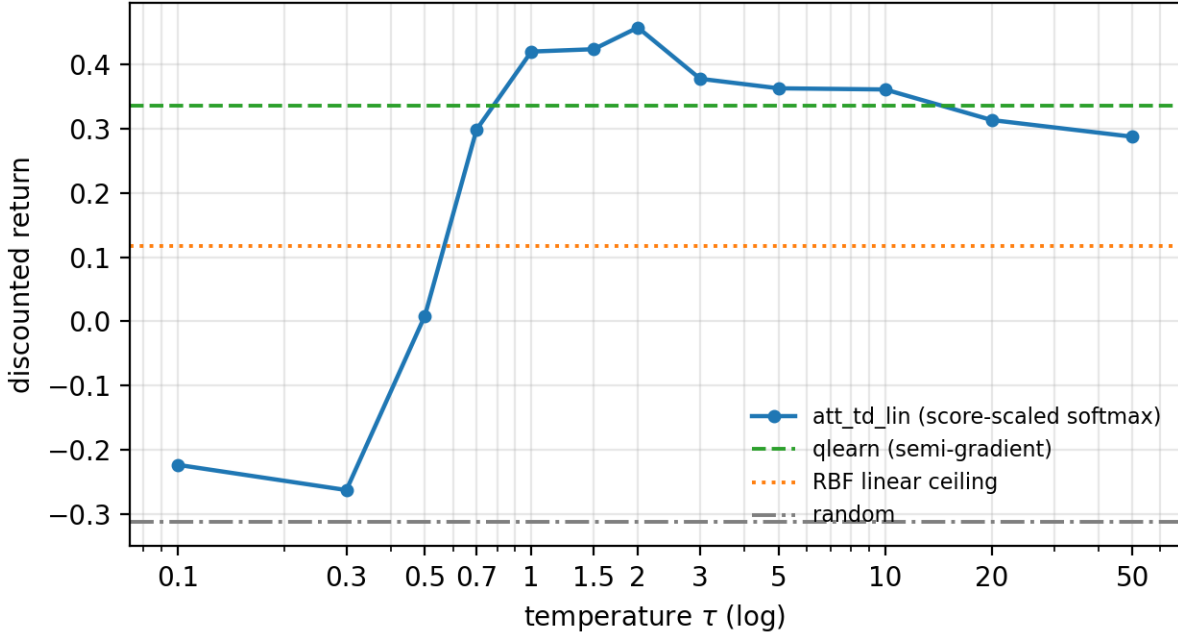


Figure 3. Temperature phase transition on sparse grid worlds. The optimum flips to the low end: $\tau = 1$ – 2 reaches 0.42–0.46, exceeding qlearn (0.34) by up to 36%, while $\tau = 0.3$ collapses below random and high temperatures fall back. The RBF linear ceiling (least-squares fit of Q^*) is low here (0.12), so it is not a meaningful upper bound for the TD-learned policy.

5.8 Mechanism attribution: smoothness $\times$ geometric alignment

Which ingredient makes low temperature a signal extractor rather than a variance trap? We decompose the variables with controlled comparisons on random tabular MDPs and grid worlds. Sparse rewards alone, determinism alone, and goal-shaped rewards alone each fail to revive $\tau = 1$ on random MDPs. The decisive variable is the feature geometry. On the same deterministic sparse grid world, replacing the RBF features by random features flips $\tau = 1$ from working (0.38) to collapsing below random (−0.38). Randomising the position mapping used by the RBF kernel (so the features remain smooth but no longer aligned with the grid adjacency) flips $\tau = 1$ from exceeding qlearn on every seed (0.33–0.45) to 0.07–0.21 (3 seeds), far below the aligned regime, and simultaneously collapses the qlearn expression ceiling (0.06–0.13 vs 0.27–0.46 aligned). An artificially smooth but unaligned RBF on random MDPs does not revive low temperature ($0.29 < \text{qlearn } 0.38$). Low-temperature one-hot focusing is therefore useful exactly when the features are

both smooth and geometrically aligned with the transition structure: $Q = \psi(s)^\top w$ is then spatially smooth, the TD error field has spatial structure concentrated near the goal, and the strongest- δ token is stable.

Direct probes of the δ field confirm this mechanism. Aligned RBF features produce a spatially smooth steady-state δ field (adjacent-state max- δ differences 0.64–0.69, vs 1.02–1.10 for permuted and 0.92–1.06 for random features). During early training, when $Q \approx 0$ and $\delta \approx r$, the δ field is strongly concentrated near the goal (Spearman $\rho \approx +0.75$ –0.79 between max- δ and inverse distance-to-goal at steps 50–250), then drifts to $\rho \approx +0.1$ –0.3 once Q is learned and δ becomes a Bellman residual; with random features ρ is negative throughout (−0.15 to −0.55), so the one-hot focus is misdirected and $\tau = 1$ collapses. Geometric alignment simultaneously guarantees the early focus (Q is smoothly learnable) and the steady-state smoothness of the δ field.

6. Discussion

Mechanism. A pure convex-combination update (softmax weights non-negative and summing to one) loses the sign of δ and deadlocks; multiplying δ back into the weights restores the sign, and in the $\tau \rightarrow \infty$ limit the operator reduces exactly to semi-gradient learning. The boundedness of closed-loop gains is consistent with the two structural obstacles identified in Section 3 — the convex hull [13] and the degradation of softmax sharpness with sequence length [14]. The temperature is not a monotone dial: it acts as a signal–noise matching filter whose optimum flips with reward sparsity (Section 5.7), and low-temperature focusing is useful only under smooth, geometrically aligned features (Section 5.8). The one structurally fatal operator remains the unscaled pure convex combination.

Finite-temperature and closed-loop scope. The large-temperature limit is strengthened by a non-asymptotic $O(1/\tau)$ deviation bound, a controlled $O(1/\tau^2)$ remainder, and a sufficient same-direction condition (Theorem 3.2, Lemma 3.3, Corollary 3.4). For one-hot features in the finite tabular full-coverage mean setting, score scaling is a bounded perturbation of the relaxed Bellman optimality update: fixed temperature yields an $O(1/\tau)$ neighbourhood and a summably annealed temperature recovers Q -learning convergence (Theorem 3.7). This is a conditional theorem, not a convergence claim for the finite-batch, constant-step-size, or function-approximation experiments reported above.

Connection to prior work. The paper fills the gap between linear-attention policy improvement [4] and softmax-attention policy evaluation [5]. The score-scaled operator is the softmax-side analogue of the semi-gradient SARSA / Q -learning update that linear attention realises exactly [4]; its large- τ limit is exactly that update. The self-containment floor is the closest softmax-side analogue to covariance-readout analyses of attention outputs [15] and to the weight-learning results established for linear attention [16].

Limits and future work. We have proved the first-order expansion, finite-temperature deviation and remainder bounds, a geometric signed-update obstruction, and a conditional tabular closed-loop guarantee. The exact $R^2 \approx 0.92$ reconstruction floor remains measured rather than proved. The tabular guarantee does not cover arbitrary linear or neural function approximation; extending

it requires a separately verified projected Bellman contraction and projection-error control. Closed-loop improvement is demonstrated experimentally on dense-reward random tabular MDPs and deterministic grid worlds with smooth, aligned features; the low-temperature regime requires such aligned features to be useful (Section 5.8). A tighter theoretical account of the phase-transition critical temperature τ_c and of the alignment conditions would also be valuable.

7. Conclusion

We asked whether a single fixed-weight softmax attention layer can perform signed, Q-learning-style policy improvement in-context, and the precise boundary of what it can express. The answer is yes, up to a single structural boundary.

We gave the first constructive demonstration that softmax attention implements policy improvement, not just evaluation. A single layer with a linear readout achieves closed-loop returns of 0.42–0.45 on random tabular MDPs, matching the numpy semi-gradient Q-learning upper bound (0.428–0.461) and exceeding one-step greedy behaviour (0.376); the mechanism transfers to deterministic grid worlds with smooth features, where a real trained softmax block reaches the linear ceiling across seeds. The improvement operator can live inside the attention weights as a score-scaled softmax, $\sum_t \delta_t \cdot \text{softmax}(\delta_t/\tau) \cdot \phi_t$, which degenerates exactly to semi-gradient learning as $\tau \rightarrow \infty$ and is itself tunable: the optimal temperature flips with reward sparsity, and low-temperature focusing is useful exactly when the features are smooth and geometrically aligned with the transition structure. The boundary is the exact self-containment of the current value: a softmax block cannot re-derive $Q = \phi^\top w$, which normalisation constrains to a reconstruction floor of $R^2 \approx 0.92$. On random tabular MDPs, and for the learned linear readout in general, every mechanism that closes the loop requires a live $\phi^\top w$ supplied externally; the sole exception we found is the fixed-formula score-scaled readout on reward-anchored grid worlds, which closes the loop even from the layer’s own evaluated memory — on random MDPs the same readout collapses under self-evaluation (0.15–0.17), so self-containment fails exactly when the value memory cannot be anchored to observed rewards, not for every readout.

The general lesson is that softmax normalisation is not the obstacle to signed improvement — the unscaled convex combination is. A signed, linearly scaled carrier (a readout, a gate, or the multiplied-back TD error) restores what the probability simplex would otherwise discard. The finite-temperature operator is quantitatively controlled: it differs from semi-gradient learning by $O(1/\tau)$, has a bounded second-order correction, and preserves the semi-gradient direction whenever that signal exceeds the stated perturbation. In the finite tabular full-coverage mean regime, this yields an $O(1/\tau)$ Bellman and greedy-policy guarantee at fixed temperature; a summably annealed temperature recovers convergence to Q^* . These claims do not extend without additional assumptions to function approximation, constant step sizes, or finite replay batches. The residual limitation is evaluation, not improvement: when value rows contain no live w , a single softmax block cannot self-contain the bilinear value estimate needed by the generic closed loop.

Future work includes extending the conditional tabular guarantee to specific projected Bellman settings, proving the exact value of the $R^2 \approx 0.92$ reconstruction floor, and deriving the phase-transition critical temperature and alignment conditions under which low-temperature focusing

helps.

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Appendix A. Reproduction commands

```
# C3-v3 closed-loop main result (random tabular MDPs)
python evaluate_c3_v3.py --ckpt results/out_c3_v3/model.pt --self-eval 0 --update-steps 400
# Score-scaled operator / gates (random tabular MDPs)
python diag_attlin.py
# Grid-world generalisation (dense/sparse, must use --reset 1)
python diag_gridworld.py --feat rbf --dense --reset 1 --eps 0.3 --update-steps 2500 --n-mdps 40
python diag_gridworld.py --feat rbf --reset 1 --eps 0.3 --update-steps 2500 --n-mdps 40
# Real softmax block on grid worlds (external  $\phi \cdot w$  vs self-evaluation)
python train_c3_v3_gridworld.py --dense --outdir results/out_c3_v3_gw
python evaluate_c3_v3_gridworld.py --ckpt results/out_c3_v3_gw/model.pt --dense --self-eval 0
python evaluate_c3_v3_gridworld.py --ckpt results/out_c3_v3_gw/model.pt --dense --self-eval 1
# Temperature phase transition (dense/sparse)
python diag_gridworld_tauscan.py --dense --eps 0.3 --update-steps 2500 --n-mdps 40 --seed 0
python diag_gridworld_tauscan.py --eps 0.3 --update-steps 2500 --n-mdps 40 --seed 0
# Alignment attribution + delta-field probes
python diag_gridworld_featalign.py --eps 0.3 --update-steps 2500 --n-mdps 40 --seed 0
python diag_gridworld_deltaprobe.py --n-mdps 15 --seed 0
python diag_gridworld_deltaevolve.py --seed 0
# Figures (regenerate from the saved JSON above; no re-running of experiments)
python tools/make_figures.py
```